Vacuum Polarization in 1+1 dimensions

Authors

Abstract

In this article...

Contents

1	Introduction	1
2	Set up	2
2.1	Field modes	2
2.2	External Field Approximation and Critical Field Strength	4
3	Self-consistent fields	5
4	Results	6
4.1	Mode sum formula	6
5	Discussion and outlook	6
A	Computing the vacuum polarization	6

1 Introduction

Vacuum polarization is a phenomenon in quantum electrodynamics (QED) by which the vacuum of a charged field polarizes when subject to external electric fields, effectively behaving like a dielectric with $\epsilon > 0$. This mechanism was one of the first QED effects studied [5], with consequences that followed closely [16; 8]. This mechanism has been already observed, e.g. in corrections to atomic levels [11] (cf. Ref. [12] for a review), while others e.g. the Schwinger effect [14] lay beyond current technological advances (cf. Ref. [6]). Particularly, interest in the Schwinger model has seen an increase in the context of lattice gauge theory, as it is one of the simplest models of a gauge theory that one can simulate, thus becoming a playground for testing novel analytical and numerical methods [2; 3].

Defining vacuum polarization for general backgrounds[1] is no easy task. Historically, [5] first defined it by assuming that the short-distance behavior of a class of states when background electric fields are present is the same, modulo smooth coefficients, as in the no background field case. In modern bibliography, these states are called Hadamard, see e.g. [7], and are particularly interesting in the context of quantum field theory on curved spacetimes, where the same idea is used to define observables necessary for the study of e.g. perturbative interacting field theory or semi-classical gravity (cf. [10; 9] for detailed discussions on the definition on such observables).

Semi-classical gravity, a possible first step towards a quantum theory of gravity, couples classical gravity to the expectation value of the stress-energy tensor, which as stated above needs to be properly defined. To our knowledge, only perturbative corrections to the metric have been studied [15]. In this article we study, as a toy model, the effect of the backreaction of a charged scalar field to a background classical electric field in 1+1 dimensional Minkowski spacetime. This setup was already studied in [1], where it was found that ignoring the backreaction of the scalar field leads to instabilities, and it is claimed that its inclusion avoids them. However, as pointed out in [13], the expression for ρ used in [1] cannot be derived from a gauge invariant prescription. Thus, the question arises: Does using the correct renormalization prescription also avoid these instabilities? We will use the prescription described in [18] to implement an iterative method to study the effect of the backreaction of the scalar field. It has been found that this expression also alleviates the above mentioned instabilities.

The layout of this article is as follows: Section 2 ...

[1] And other observables relying on the evaluation of products of fields at coinciding points.

Maybe add?: [4] discusses hadamard states in the pressence of time-like boundaries.

This article summarizes and extends the results of the M.Sc. thesis of Santiago Sanz-Wuhl, written in 2025 at the Institut für Theoretische Physik, Universität Leipzig, under the supervision of Jochen Zahn.

2 Set up

In 1+1 dimensional spacetime, two charges $q, -q$ are set at $x^1 = 0, x^1 = a$, respectively. Ignoring vacuum polarization effects, this yields a constant electric field of strength q pointing towards positive x^1 in the region $x^1 \in [0, 1]$, 0 everywhere else. We study the solutions to the semiclassical Klein-Gordon-Maxwell equations, i.e. the system of coupled differential equations

$$\left(D_\mu D^\mu + m^2\right)\phi = 0, \tag{1a}$$

$$\partial_\mu F^{\mu\nu} = \langle j^\nu \rangle, \tag{1b}$$

in the region $x = (x^0, x^1) \in \mathbb{R} \times [0, a]$. Here, $D_\mu = \partial_\mu + ieA_\mu$ is the gauge covariant derivative,

$$j^\nu(x) = \frac{ie}{2}\left((D^\nu\phi^*(x))\phi(x) - \phi^*(x)D^\nu\phi(x)\right) \tag{2}$$

is the charge current density of the Klein-Gordon field, and $\langle \mathcal{O} \rangle$ is the vacuum expectation value of an observable $\mathcal{O}$. Following [1; 18], the metric signature is $(+, -)$. The solutions looked for are the Maxwell tensor $F^{\mu\nu}$ and the two-point function of the vacuum state.

In 1+1 dimensions, A_1 can be arbitrarily set to 0. This way, Eq. (1b) reduces to the Poisson equation for the electrostatic potential A_0

$$\partial_1^2 A_0(x^1) = -\left\langle j^0 \right\rangle \equiv -\rho, \tag{3}$$

with ρ the vacuum polarization, i.e. the vacuum expectation value of the zeroth component of the charge current density. Eq. (3) will be solved with the boundary conditions

$$\partial_1 A_0(x^1)\big|_{x^1=0,a} = -q. \tag{4}$$

This way, measuring the electric field close enough to the boundary is not affected by vacuum polarization. The boundary conditions (4) fix the solution to Eq. (3) up to an addition constant, which is for convenience chosen so that $A_0(\frac{a}{2}) = 0$. With these boundary conditions and for a given $\rho(x^1)$, Eq. (3) is solved by

$$A_0(x^1) = -q\left(x^1 - \frac{a}{2}\right) - \int_{\frac{a}{2}}^{x^1}\int_0^\chi \rho(\xi)d\xi d\chi. \tag{5}$$

2.1 Field modes

In this gauge, the Klein-Gordon equation is of the form

$$\left[(\partial_0 + ieA_0)^2 + \partial_1^2 - m^2\right]\phi = 0, \tag{6}$$

which we will study using Dirichlet boundary conditions (DBC) $\phi|_{x^1=0,a} = 0$ or Neumman boundary conditions (NBC) $\partial_1\phi|_{x^1=0,a} = 0$. More general boundary conditions may be considered, though they are outside of the scope of this work. Linearity in Eq. (6) hints us towards the single frequency ansatz $\phi_n(x) = \phi_n(x^1)e^{-i\Omega_n x^0}$ with which we split the PDE into ordinary differential equations for each of the modes

$$\left[(\Omega_n - eA_0)^2 + \partial_1^2 - m^2\right]\phi_n = 0, \tag{7}$$

where the frequencies Ω_n are gauge dependent and to be determined by the boundary conditions.

The following dimensionless parameters are defined

$$z = a^{-1}x^1 \tag{8a}$$

$$\lambda = eEa^2 \tag{8b}$$

$$\varepsilon = ae \tag{8c}$$

$$\omega_n = a\Omega_{n,\alpha} + \lambda\alpha \tag{8d}$$

where $\alpha \in \mathbb{R}$ is a parameter characterizing global gauge transformations of the form

$$\varepsilon A_0 \to \varepsilon A_0 + \lambda\alpha. \tag{9}$$

This ensures the mode energies ω_n to be gauge invariant, and Eq. (7) takes the form

$$\left[(\omega_n - \varepsilon A_0)^2 + \frac{d^2}{dz^2} - a^2m^2\right]\phi_n = 0. \tag{10}$$

The modes ϕ_n are normalized with respect to the symplectic norm

$$(\phi_n, \phi_m) = i\int_0^1 (\phi_n^*\pi_m^* - \pi_n\phi_m)dz, \tag{11}$$

with $\pi_n = (D_0\phi_n)^*$ the canonical conjugate of ϕ_n.

The general solution to Eq. (6) is thus decomposed in modes

$$\phi(t,z) = \sum_{n>0} a_n\phi_n(z)e^{-i\omega_n t} + \sum_{n<0} b_n^\dagger\phi_n(z)e^{-i\omega_n t}. \tag{12}$$

Positive (negative) values of n are identified with modes of positive (negative) norm. a_n and b_n are the ladder operators for the positive and negative norm solutions, respectively, obeying canonical commutation relations $[a_n, a_n^\dagger] = [b_n, b_n^\dagger] = \delta_{nm}$, and all other commutators vanishing. The ladder operators define the vacuum state $|0\rangle$ as that for which $a|0\rangle = b|0\rangle = 0$.

As mentioned above, the definition of operators such as ρ is problematic, since it involves the product of local observables evaluated at coinciding spacetime points (see Eq. (2)). Historically, the definition of such observables was first done by Dirac by assuming a certain behavior of the vacuum at close distances. In modern literature, states obeying this condition are said to be Hadamard. Interest in Hadamard states has seen an increase in the last years in the community of Quantum Field Theory in Curved Spacetimes (cf. e.g. [17]), where the definition of observables such as the stress-energy tensor is a necessary building block for studies of semi-classical gravity or interacting fields. Hadamard states of a field ϕ are states that have a similar short-distance behavior as the vacuum of the field in the absence of background gauge fields. It is generally accepted that all physically relevant states Ω are of this form, where the short-distance divergence structure of the field is covariant, local, state independenent, and constructed from the background, i.e. the gauge connection.

Equivalently, one says that the two point function $w_\Omega^{\phi\phi^*}(x,x')$ of the scalar field ϕ in a Hadamard state Ω acquires the form

$$w_\Omega^{\phi\phi^*}(x,x') \equiv H^{\phi\phi^*}(x,x') + R_\Omega^{\phi\phi^*}(x,x'). \tag{13}$$

Here,

$$H^{\phi\phi^*} = -\frac{V_0(x,x')}{4\pi}\ln[(t-t')^2-(z-z')^2+i\varepsilon\mathrm{sign}(t-t')] \tag{14}$$

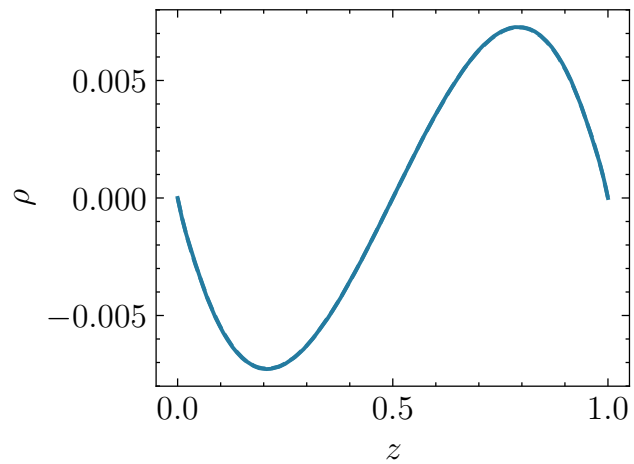

Fig. 1: The vacuum polarization ρ of the scalar field due to a background electrostatic potential $A_0(z) = -0.5(z-0.5)$, with DBC and $m = 0$.

is the Hadamard parametrix of the scalar field in 1+1 spacetime dimensions, up to order relevant for the vacuum polarization, and $R_\Omega^{\phi\phi^*}$ a state-dependent smooth function.

$$V_0(x,x') = \exp\left[-i\varepsilon\int_0^1 A_\mu(x'+s(x-x'))(x-x')^\mu ds\right] \tag{15}$$

is the parallel transport, which preserves gauge invariance in the coinciding point limit. The Hadamard condition allows us to define expectation values of local observables non-linear in the fields:

$$\left\langle D_\alpha\phi(x)D_\beta^*\phi^*(x)\right\rangle = \lim_{x'\to x} D_\alpha D_\beta^{*\prime}(w_\Omega^{\phi\phi^*}(x,x') - H^{\phi\phi^*}(x,x')), \tag{16}$$

with α, β symmetrized multiindices. The above prescription is called *point-splitting with respect to a Hadamard parametrix.*

Using these definitions, [18] calculates the expression for vacuum polarization that we will use along the paper

$$\rho = \lim_{\tau\to 0}\sum_{n>0}\Big[(\omega_n - \varepsilon A_0)|\phi_n|^2 e^{-i\omega_n\tau} + \sum_{n<0}(\omega_n - \varepsilon A_0)|\phi_n|^2 e^{i\omega_n\tau}\Big] + \frac{\varepsilon^2}{\pi}A_0. \tag{17}$$

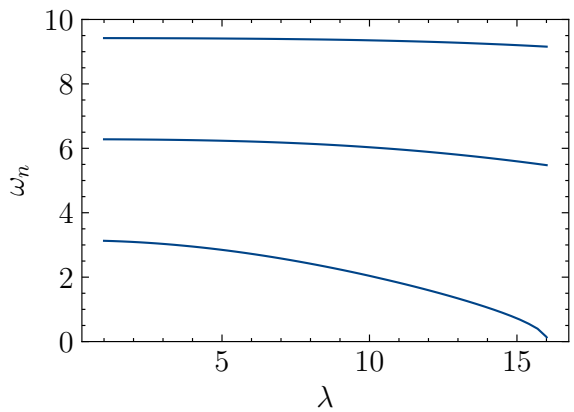

Fig. 2: Energies of the three first modes of the massless Klein-Gordon field subject to DBC as functions of λ. $\omega_1 \to 0$ as $\lambda \to \lambda_c \approx 16$ defines the critical field strength for these mass and boundary conditions configurations.

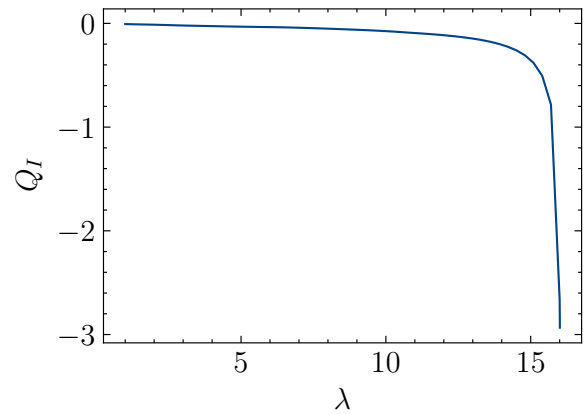

Fig. 3: Corresponding induced charge Q_I in Fig. 2 as a function of λ.

An example of ρ using this expression is given in Fig. 1, calculated in the external field approximation below explained, i.e. a homogeneous background electric field given by the potential $A_0(z) = -(z - 0.5)$.

To measure the *strength* of the vacuum polarization, the *induced charge*

$$Q_I = \int_0^{\frac{1}{2}} \rho(z) dz \tag{18}$$

is defined. This charge will screen the background electric field λ, and Gauß's law yields that the total electric field at $z = \frac{1}{2}$ is $\lambda + Q_I$.

2.2 External Field Approximation and Critical Field Strength

The external field approximation ignores the contribution of the backreaction of the charged scalar field to the background electric field. This approximation displays the instabilities of the system, which the inclusion of backreaction aims to avoid. Eq. (10) is solved with $A_0(z) = -\lambda(z - \frac{1}{2})$, and takes the form

$$\left[\left(\omega_n + \lambda\left(z - \frac{1}{2}\right)\right)^2 + \frac{d^2}{dz^2} - a^2 m^2\right]\phi_n = 0. \tag{19}$$

For each value of λ, the modes ϕ_n and their energies ω_n can be obtained from the boundary value problem. Fig. 2 displays the energies $\omega_1, \omega_2, \omega_3$ of the first three modes of the massless scalar field with DBC as functions of the background field strength λ. The value at which $\omega_1 \to 0$ defines the critical field strength $\lambda_c \approx 16.05039$. Fig. 3 displays the divergent behavior of $Q_I(\lambda)$ for this configuration as $\lambda \to \lambda_c$. λ_c adopts different values for different boundary conditions and masses.

[1] claims that the backreaction of ϕ to the background electric field avoids these instabilities. However, the expression used for the vacuum polarization cannot be derived from gauge invariant renormalization schemes; one should rather use the prescription described in [18]. This article answers the following question: does considering the backreaction using the correct prescription for the vacuum polarization avoid the instabilities that would otherwise be present?

Vacuum polarization in [1] is defined through a mode sum formula

$$\begin{aligned}\rho &= \sum_{n>0} \rho_n \\ &= \sum_{n>0} \left[(\omega_n - \varepsilon A_0)|\phi_n|^2 - (\omega_n + \varepsilon A_0)|\phi_{-n}|^2\right],\end{aligned} \tag{20}$$

so called since the total charge density is calculated as the sum of the contribution ρ_n of each mode to the to-

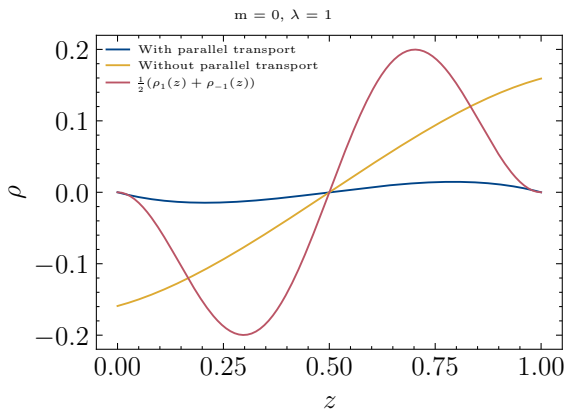

Fig. 4: Three different ρ for the same massless scalar field subject to DBC: In blue, the correct prescription as explained in [18]. In orange, ρ calculated through the mode sum (20). In red, ρ defined through the mode sum formula, truncated to the first mode.

tal charge. The above expression cannot be obtained from a gauge invariant renormalization scheme, and also relies on a specific pairing of the modes in the series to be convergent. In using the iterative procedure, [1] truncated the mode sum to the first mode. Fig. 4 shows the different ρ resulting from the different prescriptions. At first sight, it might be surprising that a field subject to DBC has non-zero vacuum polarization at the boundaries. This is precisely due to the missing $\frac{\varepsilon^2}{\pi}A_0$ term in Eq. (20). In using the iterative procedure [1] truncates the mode sum (20) to the first mode, which leads to the red curve in Fig. 4, leading to a vacuum polarization aproximately an order of magnitude stronger than the one one would obtain when using Eq. (17).

3 Self-consistent fields

The effect of the backreaction of the scalar field to the background electromagnetic field is studied by an iterative procedure. For each value of λ, at each iteration κ, one computes the following

$$\frac{d^2\phi^\kappa_{n,\lambda}}{dz^2} = -\left[(\omega^\kappa_{n,\lambda} - \varepsilon A^{\kappa-1}_{0,\lambda})^2 - a^2m^2\right]\phi^\kappa_{n,\lambda}, \tag{21a}$$

$$\begin{aligned}\rho^\kappa_\lambda = \lim_{\tau\to 0}\Big[&\sum_{n>0}(\omega^\kappa_{n,\lambda} - \varepsilon A^{\kappa-1}_{0,\lambda})|\phi^\kappa_{n,\lambda}|^2 e^{-i\omega^\kappa_{n,\lambda}\tau} \\ &+ \sum_{n<0}(\omega^\kappa_{n,\lambda} - \varepsilon A^{\kappa-1}_{0,\lambda})|\phi^\kappa_{n,\lambda}|^2 e^{i\omega^\kappa_{n,\lambda}\tau}\Big] \\ &+ \frac{\varepsilon^2}{\pi}A^{\kappa-1}_{0,\lambda},\end{aligned} \tag{21b}$$

$$\begin{aligned}A^\kappa_{0,\lambda} &= cA^{\kappa-1}_{0,\lambda} \\ &+ (1-c)\left[-\lambda\left(z-\frac{1}{2}\right) - \int_{\frac{1}{2}}^{z}\int_0^{\chi}\rho^\kappa_\lambda(\xi)d\xi d\chi\right].\end{aligned} \tag{21c}$$

This procedure has two main differences from the one used in [1]: the expression for the vacuum polarization, and the $c < 1$ prescription in Eq. (21c). The details on how ρ is calculated are explained in Appendix A. Initial conditions of this procedure are discussed below.

Noticing how Eq. (21c) acts as an update function on $A^{\kappa-1}_{0,\lambda}$, we write this scheme as an infinite dimensional fixed point problem. This is, for each λ, we look for the fixed point $A_{0,\lambda}$ obeying

$$A_{0,\lambda} = f_\lambda(A_{0,\lambda}), \tag{22}$$

with f_λ given by the composition of steps (21a), (21b) and (21c). The fixed points of f_λ, denominated "self-consistent solutions", are looked for by studying the limit $A_{0,\lambda} = \lim_{\kappa\to\infty} A^\kappa_{0,\lambda}$, where

$$A^{\kappa+1}_{0,\lambda} = f_\lambda(A^\kappa_{0,\lambda}) = f^\kappa_\lambda\left(A^0_{0,\lambda}\right), \text{with } f^\kappa_\lambda = \underbrace{f_\lambda \circ \ldots \circ f_\lambda}_{k \text{ times}}. \tag{23}$$

If the limit $A_{0,\lambda}$ exists, it is given the name self-consistent potential, as well as the corresponding self-consistent vacuum polarization ρ_λ, field modes $\phi_{n,\lambda}$ and mode energies $\omega_{n,\lambda}$.

For low enough $\lambda \ll \lambda_c$, the initial value of the sequence $A^0_{0,\lambda}$ can be set to the external field approximation $A^0_{0,\lambda}(z) = -\lambda\left(z - \frac{1}{2}\right)$. This is, however, no longer a valid initial potential for stronger background fields,

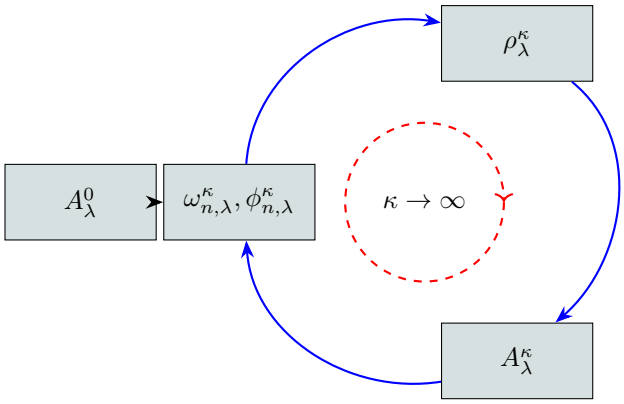

Fig. 5: Schematic of the iterative procedure for fixed λ.

as it leads to non-convergences in the sequence $A^\kappa_{0,\lambda}$. This is discussed in the Appendix ??(to be added). For higher λ-values, $A^0_{0,\lambda}$ is instead set by a stored previously calculated self-consistent potential $A_{0,\lambda'}$ with $\lambda = \lambda' + \Delta\lambda$, where $\Delta\lambda > 0$. We use this scheme to probe the λ-parameter space.

Finally, in Eq. (21c), $c \in [0, 1]$ is introduced in order to "relax" the procedure: For $c = 0$, the procedure is precisely the one used in [1], and it is expected to quickly converge. For c close to 1, updates will be small corrections to the previous potential. Stronger background field strengths increase the instability of the coupled system, thus increasing the relevance of slower converge schemes. Considering this, we adopt a scheme in which, as λ increases, c increases and simultaneously $\Delta\lambda$ decreases.

4 Results

Using the iterative procedure above laid out we study two particular cases: The massless scalar field subject to DBC, and $m = 1$ with the scalar field subject to NBC. We do not study the massless field subject to NBC, since under these boundary conditions, $\lambda_c = 0$ for $m = 0$, as pointed out in Ref. [18].

Figures 6 and 7 display the dynamics of the coupled Klein-Gordon-Maxwell system in the semiclassical approximation. The external field approximation is given for reference in orange. Notice the different λ scales. The system is studied through the energies of the first modes (top panel), the induced charge Q_I in the region $z \in [0, 0.5]$ (middle panel) and the total electric field $\lambda + Q_I$ at the midpoint $z = 0.5$ (bottom panel). We observe in both cases how considering backreaction avoids the instabilities reflected in $Q_I \to -\infty$ that the external field approximation presents. Indeed, the energy of the lower modes is raised which "prevents" ω_1 from crossing 0. We also point the reader to notice how the rapid growth of $|Q_I|$ as λ crosses λ_c keeps $\lambda + Q_I$ below λ_c.

One might also notice the almost identical behavior of the self-consistent solutions to the external field approximation in the weak-field regime $\lambda \ll \lambda_c$ in both cases.

4.1 Mode sum formula

Fig. 8 displays Q_I in the iterative procedure as a function of λ for two different prescriptions for ρ. In blue, the correct prescription used throughout this paper. In orange, ρ as defined in [1], truncated to the first mode. Notice the different behaviors in the two regimes: although the truncated mode sum formula presents a noticeably stronger vacuum polarization in the weak-field regime, their strong-field behavior is very similar. A possible explanation is the fact that the main contribution to vacuum polarization comes from the first modes, which are of course those with less energy. Thus, in strong-field regime the main contribution to (17) comes from the first modes, and rendering a truncation to first modes a very good approximation.

5 Discussion and outlook

In this article, we have studied the semiclassically coupled Klein-Gordon-Maxwell system through an interative procedure to study the effect of backreaction to a background classical electric field with a well-defined prescription of vacuum polarization.

A Computing the vacuum polarization

To be able to compute Eq. (17), some rearrangements are to be done. By use of $-\omega_n = \omega_{-n}$, ρ can be ex-

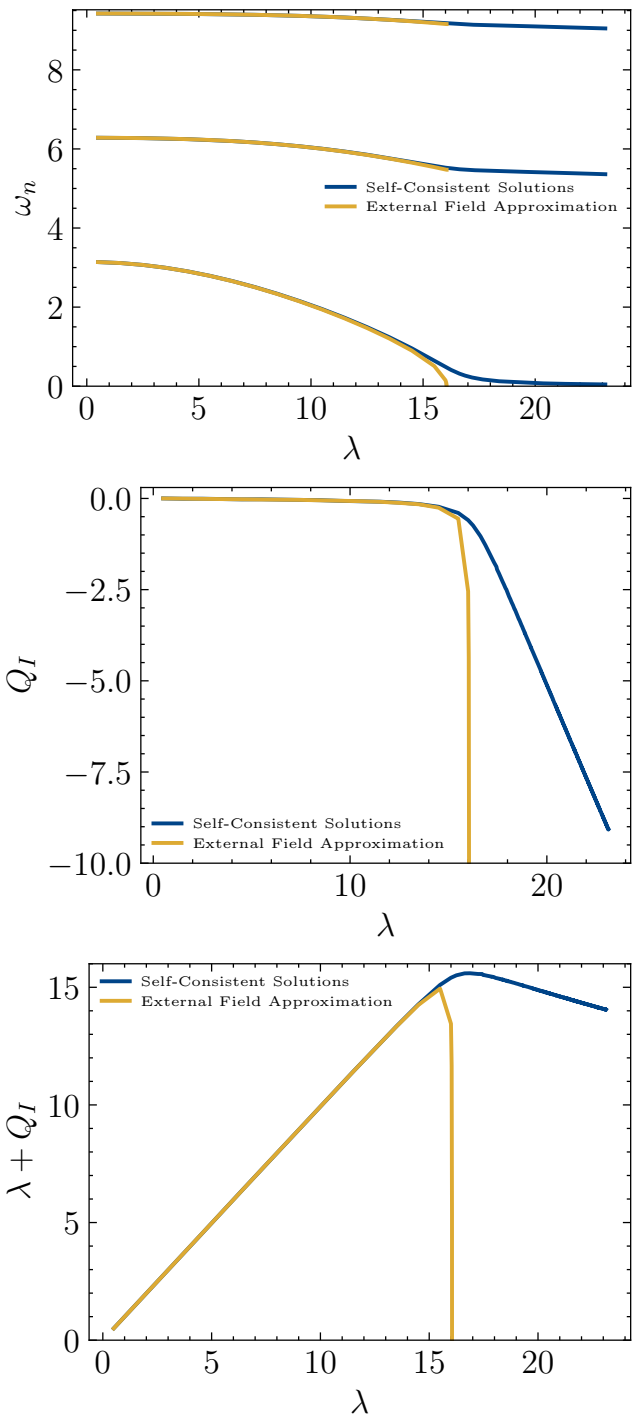

Fig. 6: In blue, the self-consistent solutions to the coupled semi-classical Klein-Gordon-Maxwell system for a massless scalar field subject to DBC. In orange, for reference, the external field approximation is given. Top panel: Energies of the first three modes as functions of λ. Middle panel: Corresponding Q_I for each λ. Bottom panel: Total electric field $\lambda + Q_I$ due to the screening of the vacuum at $z = \frac{1}{2}$.

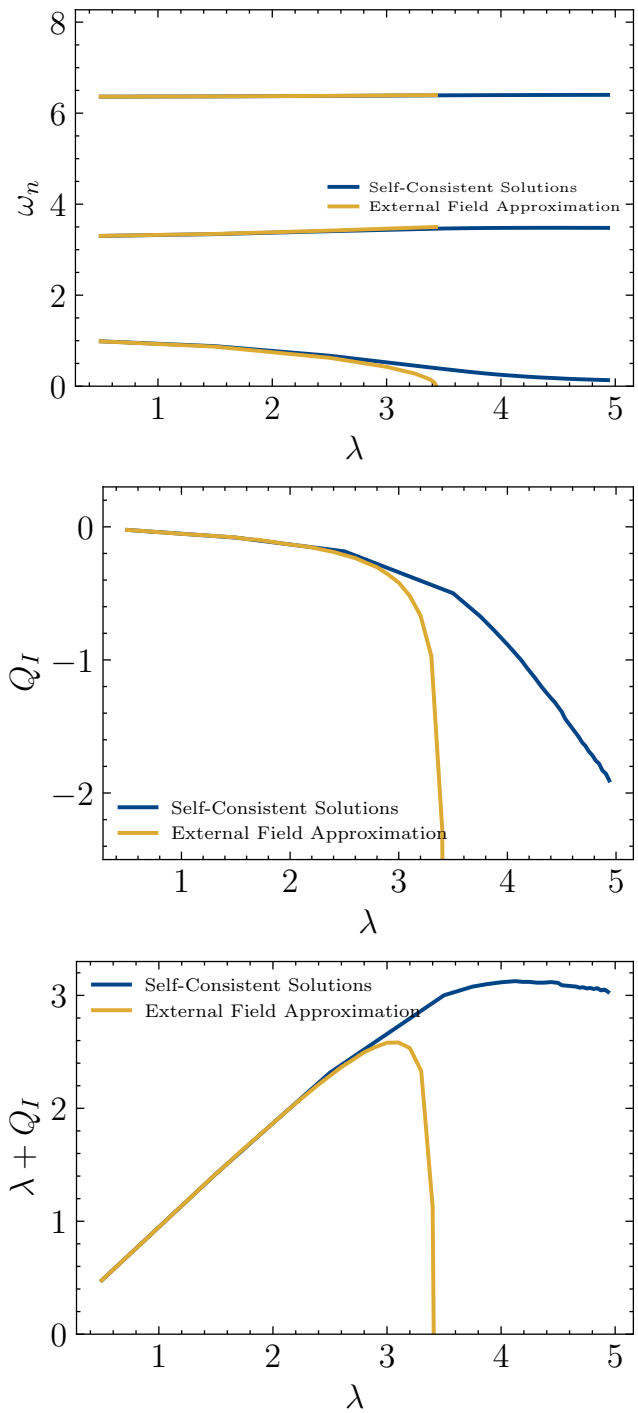

Fig. 7: Following the format in Fig. 6, the results for the unit mass scalar field subject to NBC. Further details in main text.

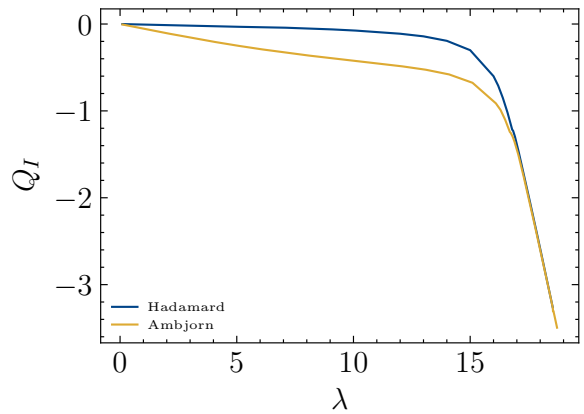

Fig. 8: Q_I for the self-consistent solutions for a massless field subject to DBC, by using different expressions for ρ. In blue, the (correct) vacuum polarization obtained through renormalization through point-splitting w.r.t. a Hadamard parametrix, and in orange the vacuum polarization obtained through a mode sum formula truncated to one mode used in [1].

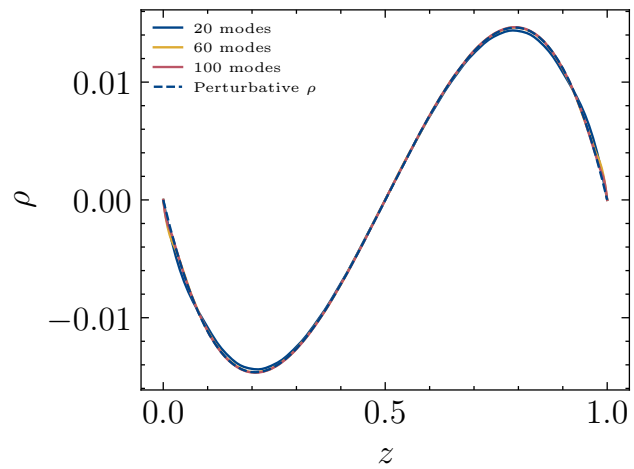

Fig. 9: ρ of the massless complex scalar field subject to a constant background electric field of strength $\lambda = 1$, for different mode cut-offs, as indicated in the legend. For reference, the perturbative vacuum polarization is given.

pressed as a single sum

$$\rho = \lim_{\tau\to 0} \sum_{n>0} \Big[(\omega_n - \varepsilon A_0)|\phi_n|^2 - (\omega_n + \varepsilon A_0)|\phi_{-n}|^2 \Big] e^{-i\omega_n \tau} + \frac{\varepsilon^2}{\pi} A_0(z) \tag{24}$$

In general, the limit and the sum do not commute. In order for them to commute, the coefficients of the complex exponentials need to decay fast enough, in order for the series to be absolutely convergent. After commuting, the sum is cut off at some finite mode N, which induces oscillations of frequency $\Delta z = \frac{1}{N+1}$. These oscillations are an artifact of the cut off, which must be smoothened out by a low-pass filter.

To achieve this, a low-pass filter is applied to the unfiltered vacuum polarization, by convoluting it with a normalized gaussian function, of approximately 3 oscillations of width. The ρ resulting from the filtering is displayed in Fig. 9, where the filtering has been applied to the vacuum polarization resulting from different mode cut-offs (20, 60 and 100, as indicated by the legend). For reference, the vacuum polarization calculated from perturbative theory as per [18] is also given.

References

[1] J. Ambjorn and S. Wolfram. Properties of the Vacuum. 2. Electrodynamic. *Annals Phys.*, 147: 33, 1983. doi: 10.1016/0003-4916(83)90066-0.

[2] M. C. Bañuls, K. Cichy, K. Jansen, and J. I. Cirac. The mass spectrum of the Schwinger model with Matrix Product States. *Journal of High Energy Physics*, 2013(11):158, Nov. 2013. ISSN 1029-8479. doi: 10.1007/JHEP11(2013)158. URL `http://arxiv.org/abs/1305.3765`. arXiv:1305.3765 [hep-lat].

[3] M. C. Bañuls, K. Cichy, J. I. Cirac, K. Jansen, and H. Saito. Thermal evolution of the Schwinger model with Matrix Product Operators. *Physical Review D*, 92(3):034519, Aug. 2015. ISSN 1550-7998, 1550-2368. doi: 10.1103/PhysRevD.92.034519. URL `http://arxiv.org/abs/1505.00279`. arXiv:1505.00279 [hep-lat].

[4] B. Costeri, C. Dappiaggi, B. A. Juárez-Aubry, and R. D. Singh. The Hadamard parametrix on half-Minkowski with Robin boundary conditions: Fundamental solutions and Hadamard states, Sept. 2025. URL http://arxiv.org/abs/2509.26035. arXiv:2509.26035 [math-ph].

[5] P. A. M. Dirac. Discussion of the infinite distribution of electrons in the theory of the positron. *Mathematical Proceedings of the Cambridge Philosophical Society*, 30(2):150–163, 1934. doi: 10.1017/S030500410001656X.

[6] A. Fedotov, A. Ilderton, F. Karbstein, B. King, D. Seipt, H. Taya, and G. Torgrimsson. Advances in QED with intense background fields. *Physics Reports*, 1010:1–138, Apr. 2023. ISSN 03701573. doi: 10.1016/j.physrep.2023.01.003. URL http://arxiv.org/abs/2203.00019. arXiv:2203.00019 [hep-ph].

[7] C. J. Fewster and R. Verch. The Necessity of the Hadamard Condition. *Class. Quant. Grav.*, 30:235027, 2013. doi: 10.1088/0264-9381/30/23/235027. _eprint: 1307.5242.

[8] W. Heisenberg and H. Euler. Consequences of Dirac's theory of positrons. *Z. Phys.*, 98(11-12): 714–732, 1936. doi: 10.1007/BF01343663. _eprint: physics/0605038.

[9] S. Hollands and R. M. Wald. Local Wick Polynomials and Time Ordered Products of Quantum Fields in Curved Spacetime. *Communications in Mathematical Physics*, 223(2):289–326, Oct. 2001. ISSN 0010-3616, 1432-0916. doi: 10.1007/s002200100540. URL http://arxiv.org/abs/gr-qc/0103074. arXiv:gr-qc/0103074.

[10] S. Hollands and R. M. Wald. Quantum fields in curved spacetime. *Phys. Rept.*, 574:1–35, 2015. doi: 10.1016/j.physrep.2015.02.001. _eprint: 1401.2026.

[11] W. E. Lamb and R. C. Retherford. Fine Structure of the Hydrogen Atom by a Microwave Method. *Phys. Rev.*, 72:241–243, 1947. doi: 10.1103/PhysRev.72.241.

[12] P. J. Mohr, G. Plunien, and G. Soff. QED corrections in heavy atoms. *Phys. Rept.*, 293(5-6):227–369, 1998. doi: 10.1016/s0370-1573(97)00046-x.

[13] J. Schlemmer and J. Zahn. The current density in quantum electrodynamics in external potentials. *Annals Phys.*, 359:31–45, 2015. doi: 10.1016/j.aop.2015.04.006. _eprint: 1501.05912.

[14] J. S. Schwinger. On gauge invariance and vacuum polarization. *Phys. Rev.*, 82:664–679, 1951. doi: 10.1103/PhysRev.82.664.

[15] J. C. Thompson and E. Winstanley. Quantum-corrected anti–de Sitter spacetime. *Physical Review D*, 110(12):125003, Dec. 2024. ISSN 2470-0010, 2470-0029. doi: 10.1103/PhysRevD.110.125003. URL https://link.aps.org/doi/10.1103/PhysRevD.110.125003.

[16] E. A. Uehling. Polarization effects in the positron theory. *Phys. Rev.*, 48:55–63, 1935. doi: 10.1103/PhysRev.48.55.

[17] R. M. Wald. The Back Reaction Effect in Particle Creation in Curved Space-Time. *Commun. Math. Phys.*, 54:1–19, 1977. doi: 10.1007/BF01609833.

[18] J. Wernersson and J. Zahn. Vacuum polarization near boundaries. *Phys. Rev. D*, 103(1):016012, 2021. doi: 10.1103/PhysRevD.103.016012. _eprint: 2010.05499.